

PROVISIONAL PATENT APPLICATION

Title of Invention

AI-Powered Grant Lifecycle Management System with Adaptive Stage Progression

Inventor Information

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Cross-Reference to Related Applications

This application claims the benefit of filing date and is entitled to the filing date of this provisional application.

BACKGROUND OF THE INVENTION

Field of the Invention

This invention relates to computerized grant application management systems, and more particularly to an intelligent, AI-powered system that guides users through a sequential grant lifecycle using adaptive stage progression based on completion thresholds.

Description of Related Art

Grant application processes are complex, time-consuming, and often overwhelming for municipal governments, nonprofit organizations, and community groups. Traditional grant management systems suffer from several critical deficiencies:

1. **Linear, Rigid Workflows:** Existing systems force users through predetermined steps regardless of their actual progress or readiness, leading to incomplete or poorly prepared applications.
2. **Lack of Intelligent Guidance:** Current systems do not provide adaptive, context-aware guidance that adjusts based on user progress and completion levels.
3. **Disconnected AI Integration:** While some systems offer AI-generated content, they lack integration with workflow progression, requiring manual coordination between content generation and application stages.
4. **No Progress-Based Gating:** Conventional systems either lock all future stages (frustrating users) or leave all stages open (resulting in premature advancement and incomplete work).
5. **Manual Stage Management:** Users must manually determine when they're ready to advance to the next stage, often advancing prematurely or delaying unnecessarily.

The current state of the art fails to provide an intelligent system that:

- Automatically assesses user readiness for stage advancement
- Uses quantifiable completion metrics to unlock subsequent workflow stages
- Integrates AI content generation as a natural part of stage progression
- Provides visual feedback on progress requirements for unlocking future stages
- Balances user autonomy with guided workflow management

Therefore, there exists a need for an improved grant lifecycle management system that intelligently guides users through sequential stages using adaptive, threshold-based progression while seamlessly integrating AI-powered content generation.

SUMMARY OF THE INVENTION

The present invention provides a novel AI-powered grant lifecycle management system that addresses the deficiencies of prior art through an intelligent, adaptive stage progression mechanism.

Primary Objectives

The primary objective of this invention is to provide a computerized system for managing grant applications through a sequential lifecycle using intelligent, threshold-based stage unlocking.

Another objective is to integrate AI content generation capabilities that activate at appropriate stages based on user progress and completion metrics.

A further objective is to provide real-time visual feedback showing users exactly what progress is required to unlock subsequent stages.

Yet another objective is to create a balanced system that prevents premature stage advancement while maintaining user control and flexibility.

Key Innovations

The invention comprises:

1. **A Five-Stage Sequential Lifecycle Model** consisting of:

- Stage 1: Discover (finding and saving grants)
- Stage 2: Evaluate (assessing fit and requirements)
- Stage 3: Prepare (gathering materials and drafting)
- Stage 4: Submit (final review and submission)
- Stage 5: Follow-up (tracking and feedback)

2. **An Adaptive Stage Unlocking Algorithm** that:

- Analyzes checklist completion percentages for each stage
- Automatically unlocks the next stage when current stage reaches 50% completion threshold
- Maintains backward accessibility to all previously unlocked stages

- Provides locked stage messaging showing exact progress requirements
3. **Integrated AI Content Generation Triggers** that:
- Activate during the "Prepare" stage when unlocked
 - Generate grant-specific sections based on municipal data and grant requirements
 - Store generated content with version tracking and word counts
 - Allow user editing while maintaining AI-generated baseline
4. **Real-Time Progress Visualization** featuring:
- Dynamic progress bar reflecting current stage position
 - Stage-specific completion badges showing tasks completed vs. total
 - Lock/unlock icons indicating stage accessibility
 - Contextual tooltips explaining unlock requirements
5. **Checklist-Based Progress Tracking** that:
- Maintains separate checklists for each lifecycle stage
 - Calculates completion percentages in real-time
 - Triggers stage transitions based on threshold achievement
 - Persists progress across user sessions

Technical Advantages

The system provides several technical advantages over prior art:

- **Intelligent Gating:** Prevents incomplete applications by requiring meaningful progress (50% threshold) before advancement
- **User Autonomy:** Allows backward navigation and re-working of completed stages without restriction
- **Scalable Architecture:** Supports unlimited concurrent grant applications with independent lifecycle tracking

- **AI Integration:** Seamlessly embeds AI generation at contextually appropriate workflow points
 - **Visual Clarity:** Provides immediate understanding of current status and advancement requirements
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DETAILED DESCRIPTION OF THE INVENTION

System Architecture

The invention comprises a web-based application built on a client-server architecture with the following components:

1. Client-Side Components (Frontend)

A. Grant Lifecycle Timeline Component

The visual interface component that displays:

- Five sequential stage indicators arranged horizontally
- Dynamic progress line connecting stages, extending based on current position
- Per-stage status icons:
 - Checkmark (✓) for completed stages
 - Alert circle for current active stage
 - Padlock icon for locked future stages
 - Empty circle for unlocked but not-yet-started stages
- Stage-specific progress badges displaying "X/Y tasks" completion ratio
- Color-coded visual states (primary blue for active/completed, gray for locked/inactive)

B. Stage Unlocking Logic

The system implements the following unlocking algorithm:

```

isStageUnlocked(stageIndex):
  IF stageIndex == 0: // First stage (Discover)
    RETURN true // Always unlocked

  IF stageIndex < currentStageIndex: // Past stages
    RETURN true // Always accessible

  IF stageIndex == currentStageIndex: // Current stage
    RETURN true // Always accessible

  IF stageIndex == currentStageIndex + 1: // Next stage
    IF currentStage == "discover":
      RETURN true // Auto-unlock Evaluate after Discover
    ELSE:
      currentProgress = getStageProgress(currentStage)
      RETURN (currentProgress.percentComplete >= 50) // 50% threshold

  RETURN false // All other future stages locked

```

C. Progress Calculation Engine

For each stage checklist:

- Tracks individual checklist items with boolean completion state
- Calculates: $\text{percentComplete} = (\text{completedItems} / \text{totalItems}) * 100$
- Updates in real-time as users complete checklist items
- Triggers UI updates when threshold (50%) is crossed

2. Server-Side Components (Backend)

A. Database Schema

Grant Applications Table:

```

grantApplications {
  id: integer (primary key)
  userId: integer (foreign key)
}

```

```
title: string
fundingAgency: string
requestedAmount: decimal
lifecycleStage: enum ['discover', 'evaluate', 'prepare', 'submit', 'follow_up']
municipality: string
grantType: string
aiGenerated: boolean
lastAiGeneration: timestamp
createdAt: timestamp
updatedAt: timestamp
}
```

Checklists Table:

```
grantChecklists {
  id: integer (primary key)
  applicationId: integer (foreign key)
  userId: integer (foreign key)
  stage: enum ['evaluate', 'prepare', 'submit', 'follow_up']
  checklistType: string
  items: jsonb (array of checklist items with completion status)
  totalItems: integer
  completedItems: integer
  percentComplete: decimal
  isComplete: boolean
  completedAt: timestamp
  createdAt: timestamp
  updatedAt: timestamp
}
```

AI-Generated Sections Table:

```
grantSections {
  id: integer (primary key)
  applicationId: integer (foreign key)
```

```
sectionName: enum ['project_goals', 'community_need', 'budget_justification', ...]
content: text
wordCount: integer
aiGenerated: boolean
createdAt: timestamp
updatedAt: timestamp
}
```

B. AI Content Generation Service

The system integrates AI (Anthropic Claude) to generate grant-specific content:

1. **Trigger Mechanism:** AI generation activates when user clicks "Generate with AI" in "Prepare" stage
2. **Context Assembly:** System gathers:
 - Municipal demographic data from government APIs
 - Grant requirements and focus areas
 - Community vulnerability indices
 - Historical grant data
3. **Section Generation:** AI produces 6+ sections:
 - Project Goals
 - Community Need Statement
 - Budget Justification
 - Project Description
 - Expected Outcomes
 - Implementation Plan
4. **Persistence:** Generated content is stored in database with metadata (word count, generation timestamp)
5. **Editability:** Users can modify AI-generated content while preserving original version

C. Lifecycle Stage Management API

Endpoint: PATCH /api/grants/applications/{id}/lifecycle-stage

Request:

json

```
{
  "lifecycleStage": "prepare",
  "notes": "Optional stage transition notes"
}
```

Process:

1. Validates requested stage is unlocked (using unlocking algorithm)
2. Updates application record with new lifecycle stage
3. Creates/updates checklist for new stage if doesn't exist
4. Returns updated application with checklist data
5. Triggers frontend UI update

Operational Workflow

User Journey Example

Step 1: Discovery Stage (Stage 1)

- User searches federal grant databases (SAM.gov, Grants.gov)
- Saves relevant grant opportunity to system
- System creates application record with lifecycleStage = "discover"
- Evaluate stage automatically unlocks (special case: no checklist in Discover)

Step 2: Evaluation Stage (Stage 2)

- User clicks on Evaluate stage (unlocked, clickable)
- System displays evaluation wizard with checklist:
 - Review eligibility requirements

- Assess organizational capacity
- Evaluate budget alignment
- Check deadline feasibility
- Rate overall fit score
- As user completes items, completedItems increments
- When percentComplete $\geq$ 50, Prepare stage unlocks
- Visual feedback: Prepare stage padlock icon changes to unlocked circle

Step 3: Preparation Stage (Stage 3)

- User clicks on Prepare stage (now unlocked)
- System displays preparation wizard with checklist:
 - Gather required documentation
 - Draft project narrative (can use AI)
 - Complete budget spreadsheet
 - Obtain letters of support
 - Review grant guidelines
- User clicks "Generate with AI" button
- System triggers AI content generation:
 - Fetches municipal data for user's jurisdiction
 - Analyzes grant requirements
 - Generates 6 grant sections
 - Stores in database
 - Displays in editable interface
- User completes $\geq$ 50% of preparation checklist

- Submit stage unlocks

Step 4: Submission Stage (Stage 4)

- User clicks on Submit stage (unlocked when Prepare $\geq 50\%$)
- System displays submission wizard:
 - Final content review
 - Required attachments checklist
 - Submission portal access
 - Confirmation of submission
- Completion of $\geq 50\%$ unlocks Follow-up stage

Step 5: Follow-up Stage (Stage 5)

- User tracks application status
- Records communications with funding agency
- Documents award/rejection outcome
- Provides feedback for future applications

[Technical Implementation Details](#)

[Frontend State Management](#)

React Query Integration:

- Queries: `['/api/grants/applications', applicationId]` - fetches application with checklists
- Mutations: Updates lifecycle stage, checklist items
- Cache invalidation: Automatically refreshes when stage changes
- Optimistic updates: UI updates immediately, rolls back on error

Component Props Flow:

GrantApplicationDetail (Parent)

```
└─ Fetches application data
└─ Passes to GrantLifecycleTimeline:
  │ └─ currentStage: string
  │ └─ checklists: GrantChecklist[]
  └─ onStageClick: (stageId) => handler
└─ Renders stage-specific wizards conditionally
```

Algorithm Complexity Analysis

Stage Unlocking: $O(1)$ constant time

- Single comparison operations
- No loops or recursive calls
- Direct array index access

Progress Calculation: $O(n)$ where n = checklist items

- Single iteration over items array
- Count of completed items
- Division operation

UI Rendering: $O(5)$ constant (5 stages)

- Fixed number of stage elements
- No dynamic rendering based on data size

Unique Technical Innovations

1. **Threshold-Based Progressive Unlock:** The 50% completion threshold represents a novel balance between:
 - Preventing premature advancement (requires meaningful progress)
 - Maintaining user momentum (doesn't require 100% completion)
 - Allowing flexibility (users can advance at 50% and return later)
2. **Bidirectional Stage Access:** Unlike traditional linear workflows:
 - Users can return to any previously unlocked stage

- Re-editing earlier stages doesn't lock future stages
- Progress is cumulative, not sequential

3. **Context-Aware AI Activation:** AI generation is:

- Stage-specific (activates in Prepare stage)
- Context-loaded (uses jurisdiction-specific data)
- Threshold-gated (only available when stage is unlocked)
- Editable post-generation (not a black box)

4. **Real-Time Visual Progress Feedback:**

- Dynamic progress line calculation: $\text{width} = (\text{currentStageIndex} / (\text{totalStages} - 1)) * 100\%$
- Contextual lock messages: Shows exact progress needed (e.g., "Complete 20% more of Evaluate to unlock")
- Status-based icon rendering (completed/current/locked states)

Scalability and Performance

Multi-Application Support:

- Each grant application has independent lifecycle tracking
- Checklist tables use applicationId foreign key for isolation
- No cross-application dependencies
- Supports unlimited concurrent applications per user

Database Optimization:

- Indexed on: userId, applicationId, lifecycleStage
- JSONB storage for checklist items (efficient queries)
- Calculated fields cached (percentComplete, totalItems)

Frontend Performance:

- Memoized calculations (stage status, unlock state)
 - Conditional rendering (only active wizard shown)
 - Lazy loading (AI content fetched on-demand)
-

CLAIMS

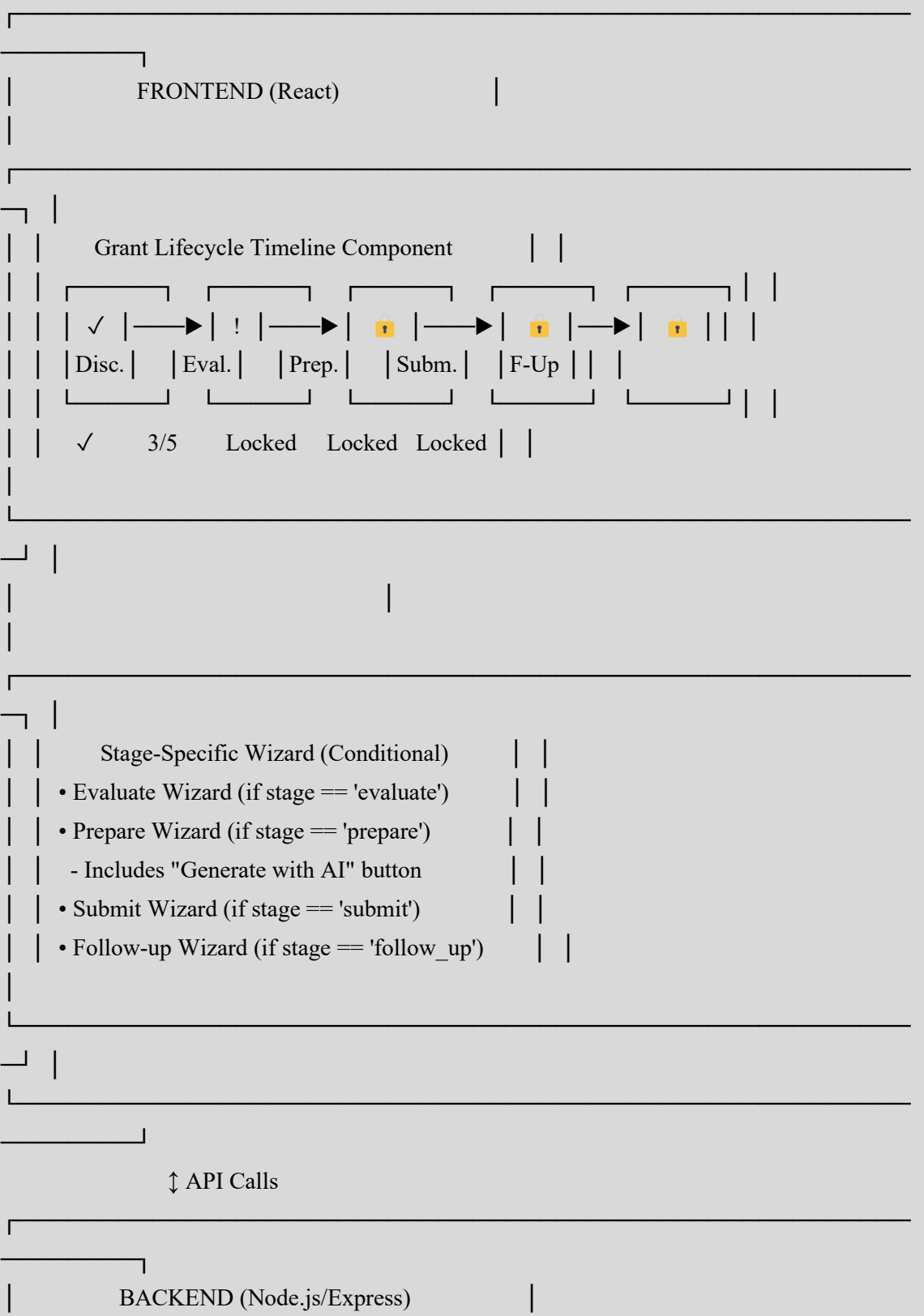
While formal claims are not required for a provisional application, the following represents the scope of the invention:

1. A computerized grant lifecycle management system comprising:
 - A sequential five-stage workflow (Discover, Evaluate, Prepare, Submit, Follow-up)
 - A stage unlocking algorithm that analyzes checklist completion percentages
 - A threshold-based progression mechanism requiring 50% completion before unlocking subsequent stages
 - An AI content generation service triggered during the Prepare stage
 - A real-time visual interface showing stage status and unlock requirements
2. The system of claim 1, wherein the stage unlocking algorithm:
 - Automatically unlocks the next sequential stage when current stage checklist completion reaches 50%
 - Maintains backward accessibility to all previously completed stages
 - Displays contextual messaging indicating exact progress required for locked stages
3. The system of claim 1, wherein the AI content generation:
 - Integrates municipal demographic data from government APIs
 - Generates multiple grant application sections simultaneously

- Stores generated content with metadata including word count and generation timestamp
 - Allows user editing of AI-generated content while preserving original version
4. The system of claim 1, wherein the visual interface:
- Displays a horizontal progress timeline with five stage indicators
 - Shows dynamic progress line extending based on current stage position
 - Renders stage-specific icons indicating completion status (checkmark, alert, lock, circle)
 - Provides per-stage completion badges showing tasks completed vs. total required
5. A method for adaptive grant application workflow management comprising:
- Tracking user progress through stage-specific checklists
 - Calculating completion percentage as $(\text{completed items} / \text{total items}) \times 100$
 - Comparing completion percentage to 50% threshold
 - Automatically unlocking next stage when threshold is met
 - Providing visual feedback on unlock requirements for locked stages
6. The method of claim 5, further comprising:
- Triggering AI content generation when Prepare stage is unlocked
 - Gathering jurisdiction-specific community data as generation context
 - Generating multiple grant sections using large language model AI
 - Storing generated content in database with version tracking
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DRAWINGS AND FIGURES

Figure 1: System Architecture Diagram



Lifecycle Stage Management API		
PATCH	/api/grants/applications/{id}/lifecycle-stage	
- Validates unlock status - Updates application stage - Creates/updates checklist		
AI Content Generation Service		
POST	/api/grants/applications/{id}/generate	
- Fetches municipal data (demographics, etc.) - Calls Anthropic Claude API - Generates 6+ grant sections - Stores in database		
Progress Tracking Service		
- Calculates completion percentages - Evaluates unlock thresholds (50%) - Updates checklist records		

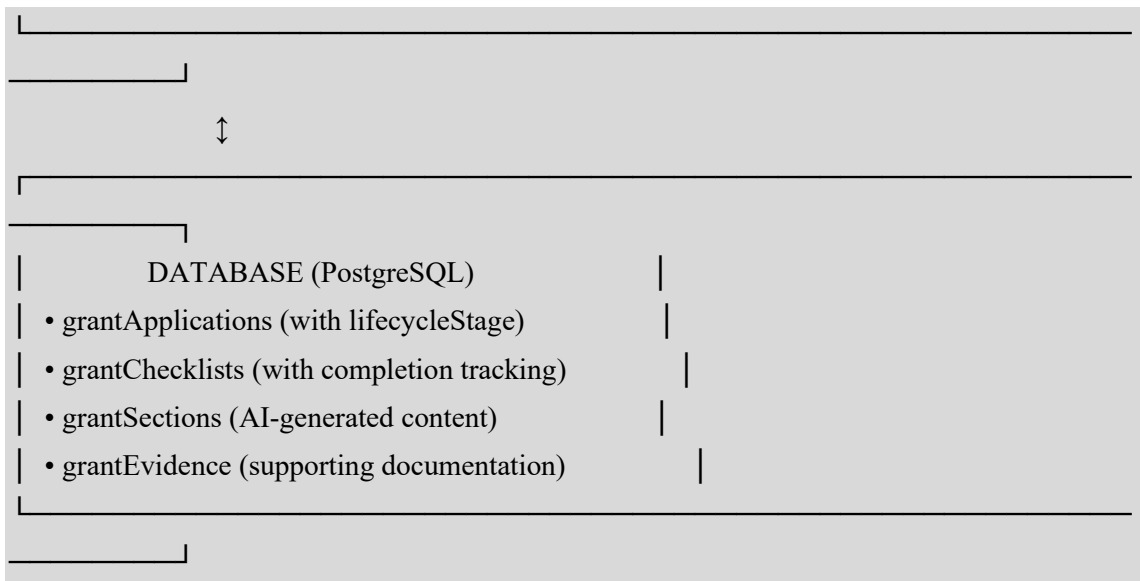
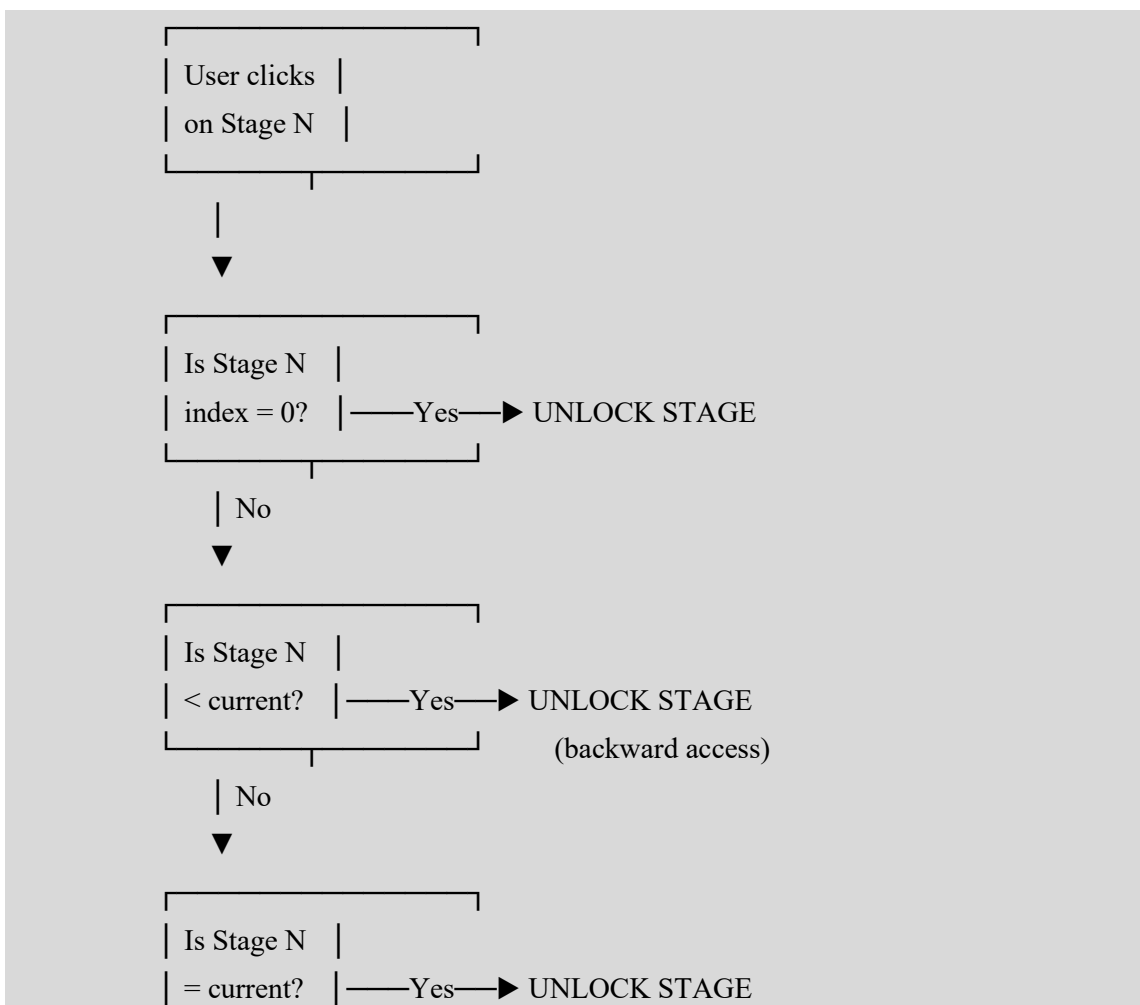


Figure 2: Stage Unlocking Algorithm Flowchart



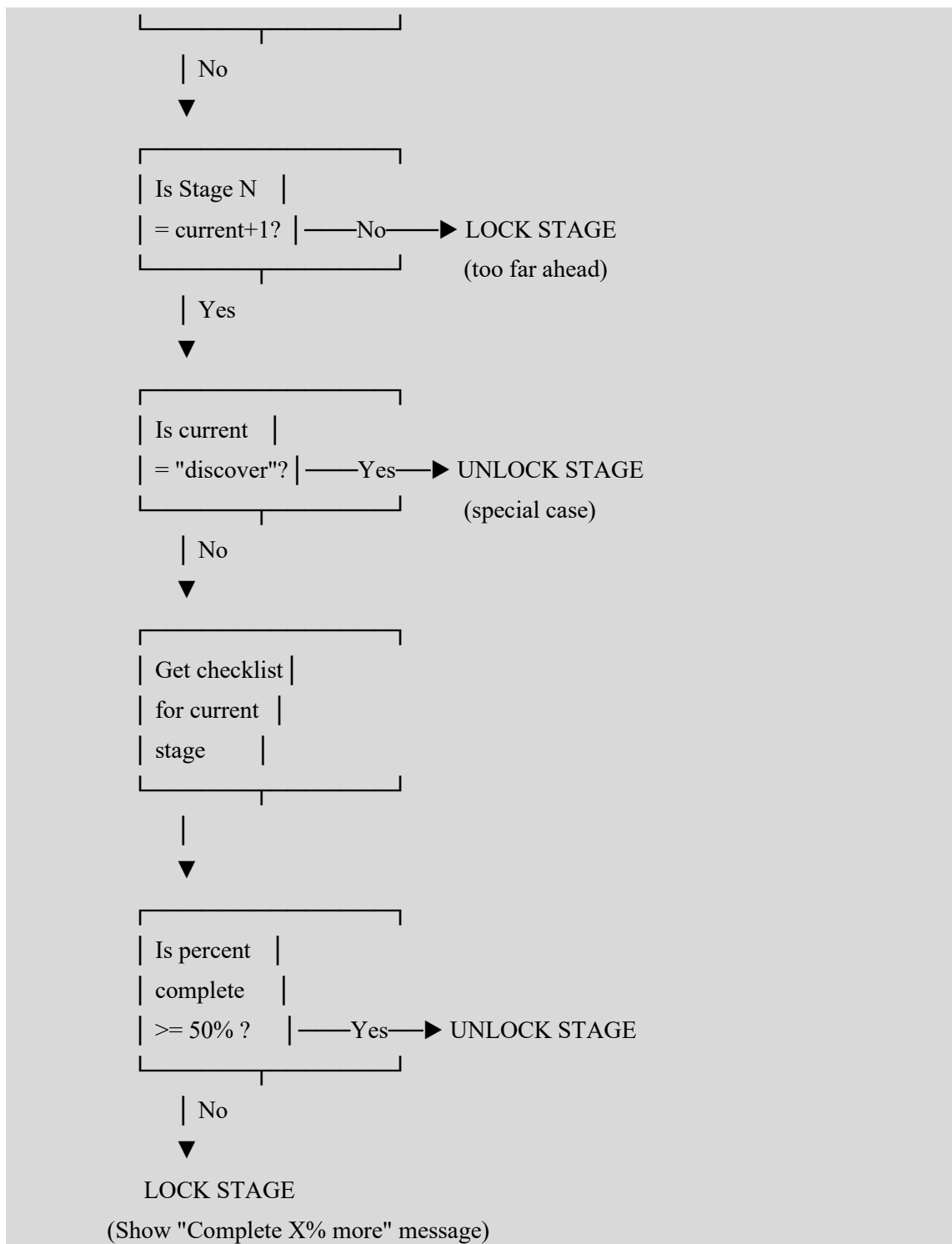
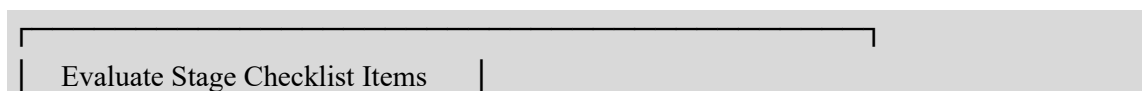


Figure 3: Checklist Completion Calculation



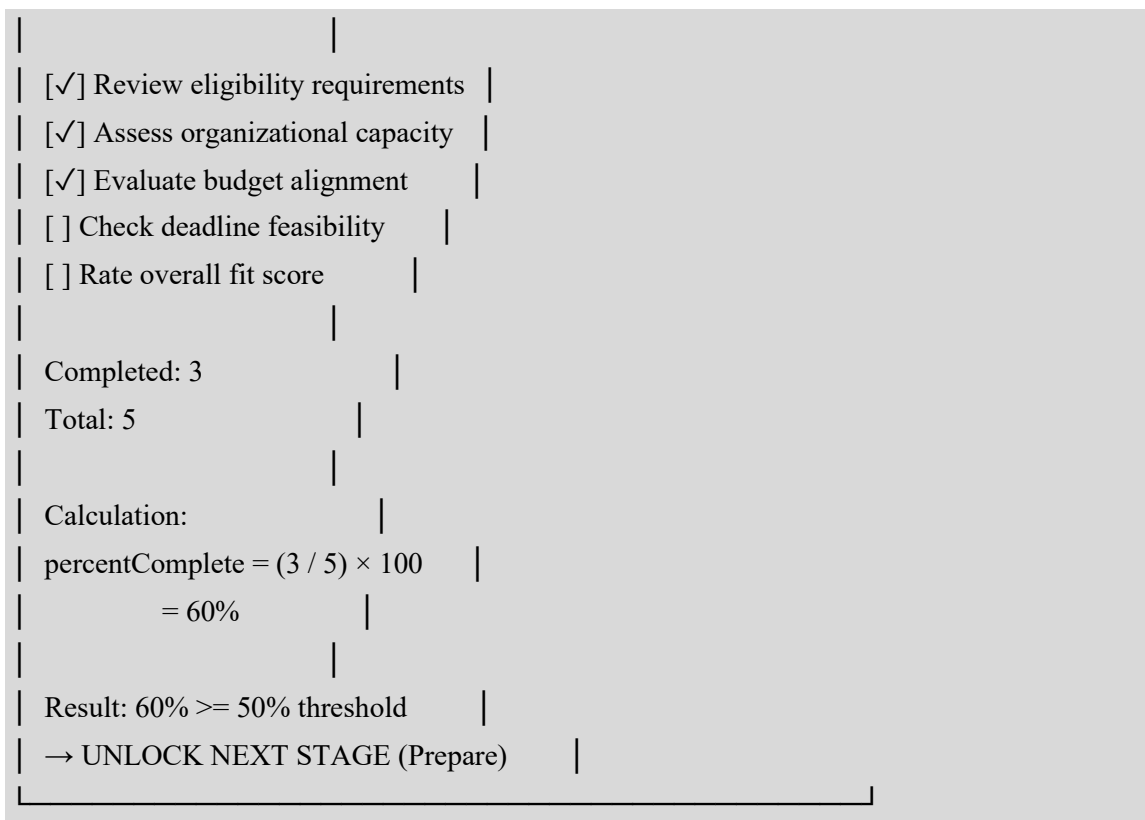
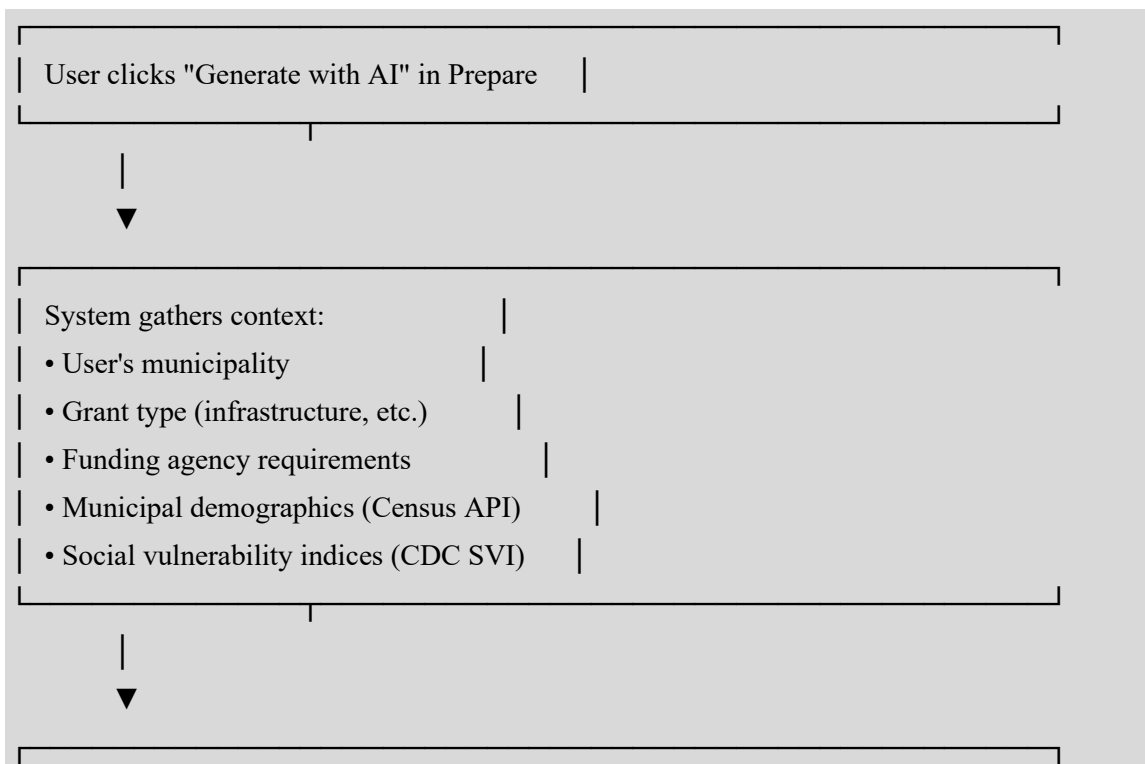
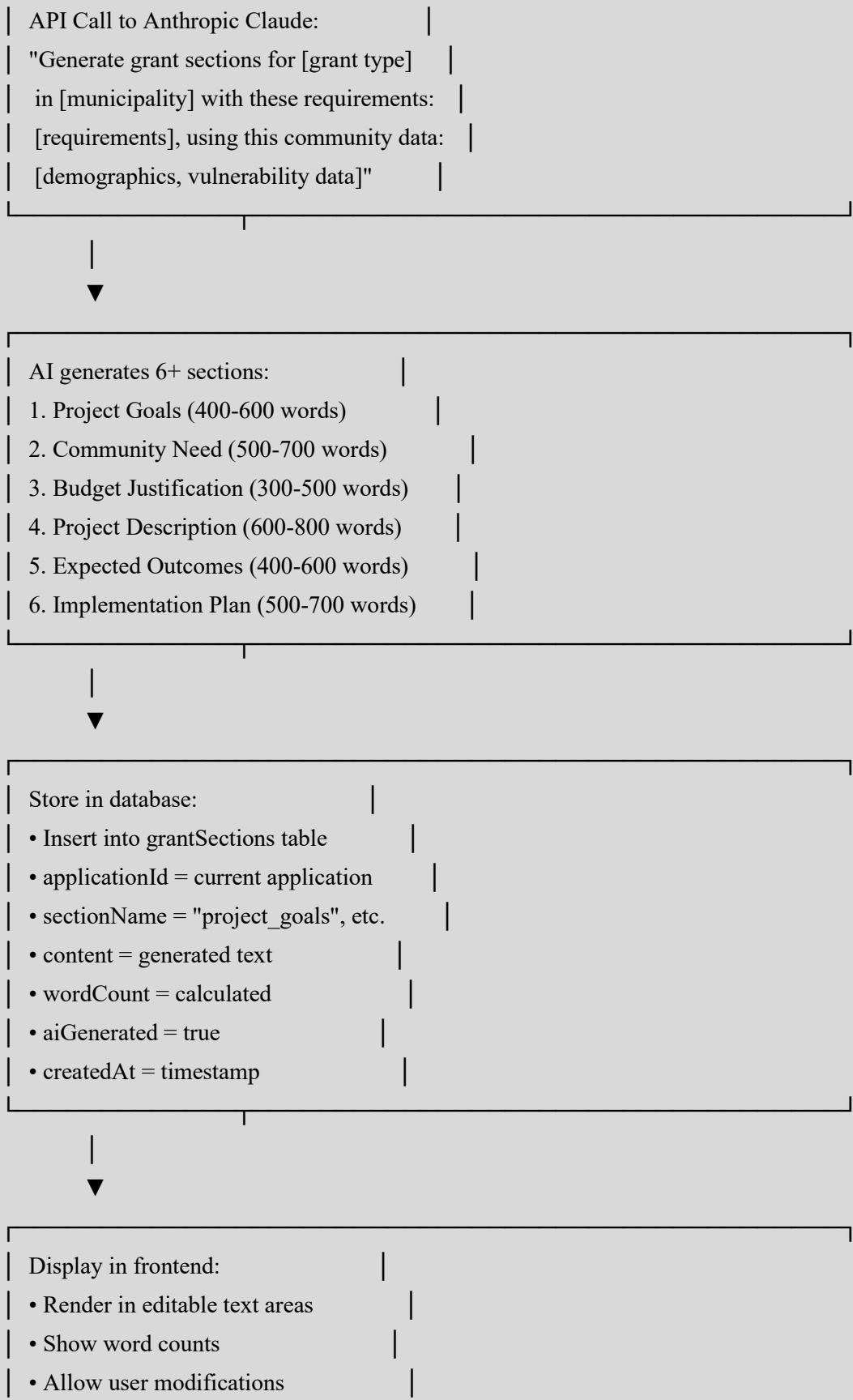


Figure 4: AI Content Generation Workflow





• Save edits back to database	
-------------------------------	--

Figure 5: Visual UI States

Stage Status: COMPLETED	
✓	Checkmark icon
Discover	Primary blue color
(completed)	Always clickable
✓ Complete	Badge showing done
Stage Status: CURRENT	
!	Alert icon
Evaluate	Primary blue with ring
(current)	Always clickable
3/5 tasks	Progress badge
Stage Status: UNLOCKED (next)	
○	Empty circle
Prepare	Gray/muted color
(unlocked)	Clickable
	No badge yet
Stage Status: LOCKED	
🔒	Lock icon
Submit	Faded gray
(locked)	Not clickable
50% opacity	Tooltip on hover:
"Complete 20% more	

ADVANTAGES OVER PRIOR ART

The present invention provides numerous advantages over existing grant management systems:

1. **Intelligent Progress Gating:** The 50% completion threshold prevents users from advancing prematurely while not requiring 100% completion, which would be overly restrictive. This balance is novel in grant management workflows.
 2. **Adaptive User Guidance:** The system automatically assesses readiness and provides specific feedback (e.g., "Complete 20% more") rather than generic "locked" messages.
 3. **Seamless AI Integration:** Unlike systems that treat AI as a separate tool, this invention integrates AI generation as a natural workflow step that activates when contextually appropriate (Prepare stage unlocked).
 4. **Bidirectional Access:** Users can return to any unlocked stage to make improvements without affecting future stage accessibility - a significant usability improvement over rigid linear systems.
 5. **Visual Clarity:** The dynamic progress bar, status icons, and contextual tooltips provide immediate understanding of current position and advancement requirements without reading documentation.
 6. **Scalability:** The architecture supports unlimited concurrent applications with independent lifecycle tracking, making it suitable for organizations managing multiple grant applications simultaneously.
 7. **Data-Driven Content:** AI generation uses real-time municipal data from authoritative government sources, ensuring grant applications contain accurate, relevant community information.
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ALTERNATIVE EMBODIMENTS

While the preferred embodiment uses a 50% completion threshold, the system can be configured with alternative thresholds (e.g., 40%, 60%, 75%) based on organizational policies or grant complexity.

The five-stage lifecycle can be extended or reduced. For example:

- Simpler grants might use: Discover → Prepare → Submit
- Complex multi-year grants might add: Planning → Discover → Evaluate → Prepare → Budget → Submit → Award → Execute → Report

The AI content generation can use different AI providers (OpenAI GPT, Google PaLM, etc.) or multiple providers in ensemble for improved quality.

The visual timeline can be rendered vertically for mobile devices or as a circular progress indicator for dashboard views.

Checklist items can be weighted (some items worth more completion percentage than others) to prioritize critical tasks.

CONCLUSION

This provisional patent application describes a novel AI-powered grant lifecycle management system that addresses critical deficiencies in existing systems through intelligent, threshold-based stage progression and seamlessly integrated AI content generation.

The invention's unique combination of adaptive unlocking algorithms, real-time progress tracking, and context-aware AI activation creates a superior user experience while ensuring grant applications are thoroughly prepared before submission.

The system's technical architecture is scalable, performant, and extensible, making it suitable for widespread adoption across municipal governments, nonprofit organizations, and community groups seeking efficient grant application management.

APPENDICES

Appendix A: Source Code References

Key implementation files from CommunityInsight.AI platform:

1. **Frontend Timeline Component:** client/src/components/grant-lifecycle-timeline.tsx
 - Lines 74-96: Stage unlocking algorithm (isStageUnlocked function)
 - Lines 84-91: 50% completion threshold logic
 - Lines 98-114: Contextual lock messaging
2. **Frontend Application Detail:** client/src/pages/grant-application-detail.tsx
 - Lines 152-179: AI content generation mutation
 - Lines 206-223: Lifecycle stage update mutation
 - Lines 124-149: Section loading and hydration
3. **Backend AI Service:** server/ai-grant-writer-service.ts
 - Lines 49-108: Main grant generation workflow
 - Lines 54-58: Community intelligence gathering
 - Lines 82-83: Section generation trigger
4. **Backend Routes:** server/routes.ts
 - Lines 3965-4047: AI content generation endpoint
 - Lines 12326-12350: Grant application generation API

Appendix B: Database Schema Details

Full table definitions with constraints, indexes, and relationships available in:

- shared/schema.ts (Drizzle ORM definitions)
- shared/grants-schema.ts (Grant-specific schemas)

Appendix C: Technical Dependencies

Core technologies used in implementation:

- Frontend: React 18, TypeScript, TanStack Query, Wouter routing
 - Backend: Node.js, Express.js, TypeScript
 - Database: PostgreSQL with Drizzle ORM
 - AI: Anthropic Claude API
 - Government Data: Census Bureau API, SAM.gov API
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FILING INFORMATION

Application Type: Provisional Patent Application

Fee Type: Micro Entity (\$65) / Small Entity (\$130)

Filing Method: USPTO Patent Center (electronic)

Estimated Filing Date: [Date of filing]

This concludes the Provisional Patent Application for "AI-Powered Grant Lifecycle Management System with Adaptive Stage Progression"